

CREST Phase 1 Technical Document

Metric-Response Validation, Reproducibility Specification, and Empirical Positioning

Framework: Alethian / Phenomenurgic Response-Field Program

Primary empirical branch: CREST galaxy-scale metric response

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Status: Phase 1A technical consolidation with regenerated reproducibility-facing assets

Abstract

This document consolidates Phase 1 of the CREST validation campaign into a formal technical specification. CREST is treated here as a narrow galaxy-scale response-field model, not as a completed replacement for dark matter, MOND, or relativistic gravity. The document freezes the operational claim, declares the invariant roles χ , η , I , F , defines the reserve and degraded-reserve constraints, states the weak-field galaxy projector, and records the existing SPARC benchmark position. The archived CREST II benchmark used the SPARC sample of 175 galaxies and 3391 radial points with fixed stellar mass-to-light ratios. In that benchmark, the fixed CREST backbone gave a grouped galaxy-held-out RMSE of 0.2039 in $\log_{10} g$, compared with 0.1996 for a fixed MOND-style branch and 0.1998 for a fixed RAR-form branch. The shared CREST Huber residual layer gave 0.1959, improving overall on the fixed MOND/RAR branches while showing its strongest gains in structurally messier $Q = 2$ and $Q = 3$ subsets. Tuned pISO and NFW halo references remained much stronger as in-sample galaxy-by-galaxy fits. The correct Phase 1 conclusion is therefore mixed but useful: the backbone alone does not beat fixed MOND/RAR; a shared residual layer can improve held-out performance; tuned halo templates still define a substantial remaining empirical gap; and the next decisive work is not more unconstrained flexibility, but an independent, non-target-derived projector for the response and disequilibrium sectors. The final sections regenerate the missing public-facing bundle as a reproducibility specification: data manifest, frozen equations, train/test protocol, result tables, figure assets, notebook skeleton, falsification criteria, and Phase 1B execution checklist.

1. Phase 1 objective

The first public task for the framework is not to prove the entire Alethian ontology. The first task is to make one narrow empirical claim precise enough to be tested, criticized, reproduced, and potentially broken. CREST is the correct first branch because it maps the invariant grammar onto public galaxy data and can be compared against existing baselines. The technical posture is deliberately conservative: CREST is a response-layer hypothesis over observed residual structure. It does not claim that dark matter is false, that MOND is defeated, or that the relativistic completion is finished. It asks whether an organized metric-response projector captures a nontrivial portion of the galaxy residual field under held-out validation.

- Freeze the galaxy-facing CREST claim before further fitting or narrative expansion.
- Reconstruct the Phase 1 benchmark tables and figures from archived CREST records.
- Declare what counts as success, partial success, and failure.
- Separate proof-of-concept empirical results from future radical or cross-domain implications.
- Produce a document suitable for technical critique by readers familiar with rotation curves, RAR/MOND comparisons, and model validation.

2. Short verdict

Phase 1 should be stated as a disciplined partial result. The existing benchmark does not support the claim that the fixed CREST backbone already outperforms simple MOND/RAR baselines. It does support a narrower claim: a shared CREST residual layer, trained under grouped galaxy-held-out validation, improves the aggregate log-acceleration RMSE relative to the fixed one-parameter comparators, with the improvement concentrated in messier quality classes. This is compatible with the intended interpretation of the response layer as a secondary sector that activates when clean circular support is no longer the whole story.

The result is not final closure. Tuned pISO and NFW halo fits remain much stronger when fit galaxy by galaxy. The Phase 1 result therefore creates a credible empirical foothold, not a full astrophysical replacement theory.

3. CREST as a response-layer model

CREST is used here as a Covariant Relativistic Extended Scalar Theory branch of the larger Alethian / Phenomenurgic response framework. The core philosophical claim is translated into operational physics as follows: observations do not directly record unconstrained hidden conditions; they record projected response. A galaxy rotation curve is therefore not treated merely as matter producing acceleration, but as baryonic structure plus response capacity, ordered attachment, structural information, and irreversible loading projecting into an accessible acceleration record.

The four role variables are not arbitrary fit labels. Their division of labor is the guardrail that prevents the framework from becoming an unlimited residual absorber.

Symbol	Operational role in Phase 1
chi	Matter-conditioned metric-response amplitude or backbone response field. In the galaxy branch, chi projects into the residual acceleration g_chi.
eta	Ordered attachment or relational coupling of the response state to the baryonic host.
I	Stored structural information: gradients, curvature, branch thickness, and organized residual shape.
F	Irreversible loading, disequilibrium broadening, non-circular contamination, transport-sensitive departure.
R	Undegraded reserve, $R = \eta I - \chi^2$. It measures support margin inside the admissible response cone.
R_F	Degraded reserve, $R_F = \eta I \exp(-\lambda_F F) - \chi^2$. It measures response capacity after irreversible loading.
S_F	Logarithmic degraded reserve field, $S_F = \ln(\eta I \exp(-\lambda_F F) / \chi^2)$. The response boundary is $S_F = 0$.

4. Frozen mathematical core

The Phase 1 document freezes the following equations as the public technical core. Later work may improve estimators, but the paper should not change these definitions ad hoc after seeing validation outcomes.

```
Undegraded reserve:
  R = eta I - chi^2

Degraded response capacity:
  C_F = eta I exp(-lambda_F F)

Degraded reserve:
  R_F = eta I exp(-lambda_F F) - chi^2

Admissibility condition:
  chi^2 <= eta I exp(-lambda_F F)

Degraded logarithmic reserve field:
  S_F = ln[ eta I exp(-lambda_F F) / chi^2 ]

Response boundary:
  S_F = 0, equivalently chi^2 = eta I exp(-lambda_F F)
```

These equations impose a capacity condition. They do not by themselves fit a rotation curve. Observable predictions require a domain projector that maps the invariant response state into a measured quantity.

5. Galaxy projector and weak-field reduction

The response tensor is the parent object. A rotation curve only sees a projected scalar record. The Phase 1 galaxy projector maps the response state into a residual acceleration channel. The weak-field galaxy-facing relation is:

Observed acceleration decomposition:

$$g_{\text{obs}}(R) = g_{\text{bar}}(R) + g_{\text{chi}}(R)$$

Fixed CREST backbone prediction:

$$g_{\text{pred},0}(R) = g_{\text{bar}}(R) + \sqrt{a_0 g_{\text{bar}}(R)}$$

Residual response interpretation:

$$g_{\text{chi}}(R) = |\text{grad } \Phi_{\text{chi}}(R)|$$

The fixed backbone is intentionally restrictive. It is not allowed to retune itself galaxy by galaxy. The residual layer must then prove that it adds organized, generalizable structure rather than absorbing targets by leakage.

6. Phase 1 data manifest

The Phase 1 validation package uses the SPARC/RAR/BTFR public galaxy records as the empirical base. Drive already contains the necessary source files and prior CREST records; the missing public-facing assets are regenerated here as document tables, figures, and a reproducibility specification.

Asset	Role	Status
SPARC_Lelli2016c.mrt.txt	Galaxy-level SPARC sample metadata and quality classes.	Located in Drive. Required for executable reproduction.
MassModels_Lelli2016c.mrt.txt	Radial rotation curve and baryonic component mass-model data.	Located in Drive. Required for executable reproduction.
RAR.mrt.txt	Radial acceleration relation comparison record.	Located in Drive. Required for RAR comparisons.
BTFR_Lelli2019.mrt.txt	Baryonic Tully-Fisher relation support record.	Located in Drive. Optional for Phase 1 context.
crest2_updated.pdf	Archived benchmark values, equations, tables, and interpretation.	Recovered from ALETHIAN archive. Used as numerical backbone.
CREST_metric_response_SPARC_paper.pdf	Earlier SPARC narrative consolidation and references.	Recovered from ALETHIAN archive. Used as supporting source.
CREST_IV_metric_response_projector.pdf	Projector discipline, degraded reserve, and falsification language.	Recovered from ALETHIAN archive. Used for technical framing.
Regenerated CSV tables	Machine-readable copies of Phase 1 benchmark values.	Regenerated as companion assets during compilation.
Regenerated figures	RMSE and baseline-reference plots.	Regenerated as companion assets during compilation.

7. Validation protocol

The benchmark protocol must be grouped by galaxy identity. Radial points from the same galaxy are not independent for model selection because they share mass-to-light assumptions, distance, inclination, surface-brightness structure, and unmodeled systematics. A fair held-out test trains on galaxies and evaluates on distinct galaxies.

- Primary target: $y_i = \log_{10}(g_{\text{obs},i})$.
- Primary metric: RMSE in $\log_{10} g$ over held-out radial points.
- Grouping: galaxy-held-out folds, preferably grouped 5-fold cross-validation.
- Baryonic construction: gas, disk, and bulge contributions with fixed stellar mass-to-light ratios unless a separate sensitivity run is declared.
- Main fixed-branch comparators: CREST backbone, fixed MOND-style simple interpolation, and RAR-form branch.
- Reference halo baselines: pISO and NFW fits are treated as tuned in-sample references unless a matched hierarchical or held-out halo protocol is implemented.
- Leakage rule: no use of $g_{\text{obs}} - g_{\text{bar}}$ as an oracle feature in a public predictive model. Diagnostic target-derived tests must be labeled as diagnostic only.

8. Baseline equations

The Phase 1 paper should keep the comparator equations explicit. The one-parameter MOND-style branch and RAR-form branch test whether CREST adds value beyond known acceleration-law behavior.

CREST backbone:

$$g_{\text{CREST0}}(R) = g_{\text{bar}}(R) + \sqrt{a_0 g_{\text{bar}}(R)}$$

MOND-style simple interpolation comparator:

$$g_{\text{MOND}}(R) = 0.5 g_{\text{bar}}(R) [1 + \sqrt{1 + 4 a_0 / g_{\text{bar}}(R)}]$$

RAR-form comparator:

$$g_{\text{RAR}}(R) = g_{\text{bar}}(R) / [1 - \exp(-\sqrt{g_{\text{bar}}(R) / a_0})]$$

Residual-layer model family:

$$y_i = \log_{10}(g_{\text{pred}}, \theta(R_i)) + \Delta_{\text{theta}}(\text{features}_i) + \epsilon_{\text{residual}_i}$$

The residual-layer model is acceptable only when its features are generated from baryonic structure, admissible response variables, or independent kinematic complexity records. It is not acceptable to build the public predictor from target-derived residuals unless the section is explicitly labeled diagnostic.

9. Reconstructed Phase 1 benchmark results

The following benchmark values are reconstructed from the archived CREST II technical paper. The SPARC sample is reported there as 175 galaxies and 3391 radial points. The fixed stellar mass-to-light assumptions are $\text{Upsilon}_{\text{d}} = 0.5$ and $\text{Upsilon}_{\text{b}} = 0.7$. The main result is mixed: the fixed CREST backbone does not beat fixed MOND/RAR, while the shared Huber residual layer improves the aggregate RMSE and performs best in $Q = 2$ and $Q = 3$ subsets.

Model	Full RMSE	Q=1	Q=2	Q=3
CREST backbone	0.2039	0.1703	0.2192	0.4518
Fixed MOND-style branch	0.1996	0.1700	0.2116	0.4329
Fixed RAR-form branch	0.1998	0.1699	0.2121	0.4338
CREST Huber residual layer	0.1959	0.1710	0.2030	0.4172

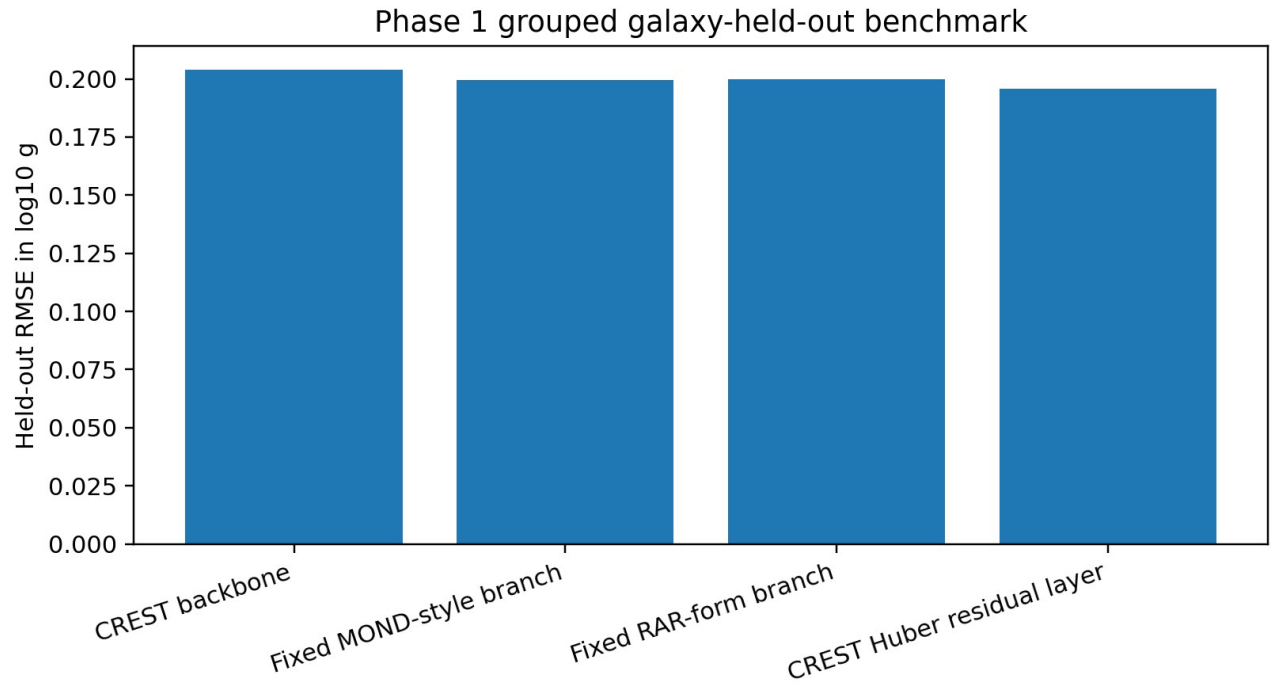


Figure 1. Full held-out RMSE in $\log_{10} g$ for the Phase 1 model hierarchy.

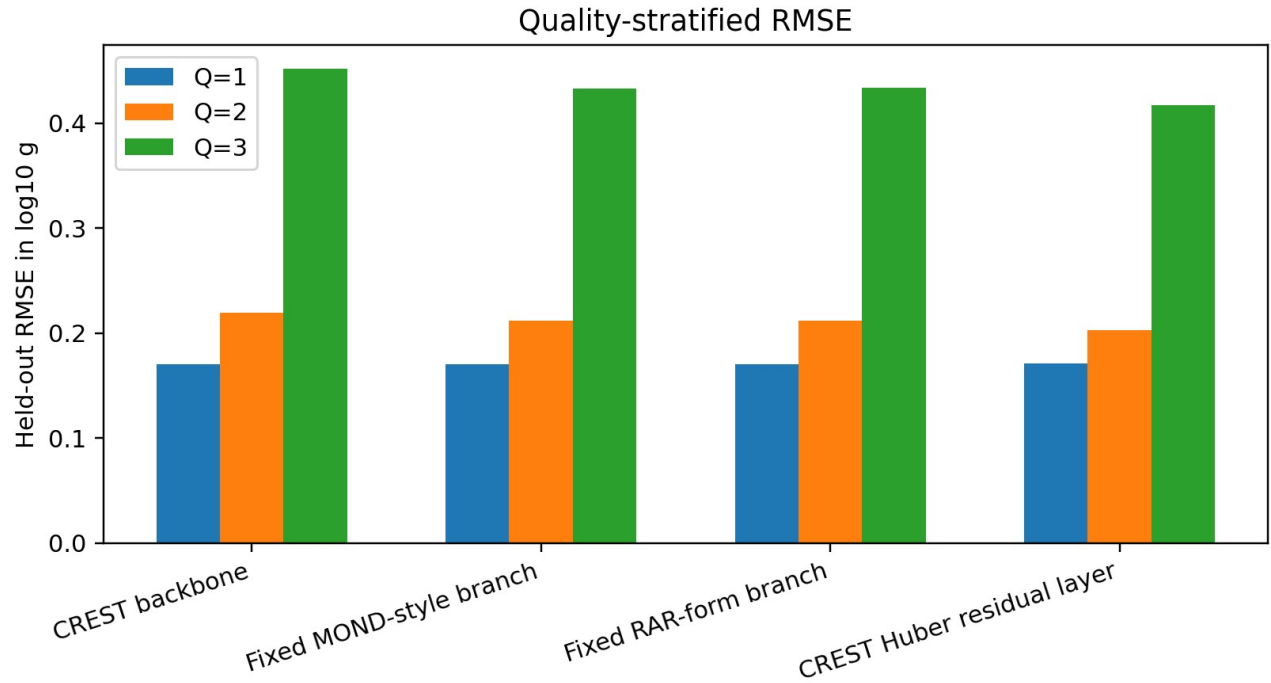


Figure 2. Quality-stratified RMSE. Lower values are better.

10. In-sample halo reference gap

The tuned halo references are not the same kind of predictive baseline as a fixed held-out law because they are fit galaxy by galaxy. They are still important because they show the empirical gap between a global response law and

flexible galaxy-specific mass modeling. The correct conclusion is not that Phase 1 has defeated halo modeling. The correct conclusion is that the residual-layer branch has a measurable foothold while a large halo-reference gap remains.

Model	Median per-galaxy RMS (km/s)
CREST backbone	12.88
Fixed MOND-style branch	12.27
Tuned pISO halo	3.42
Tuned NFW halo	4.74

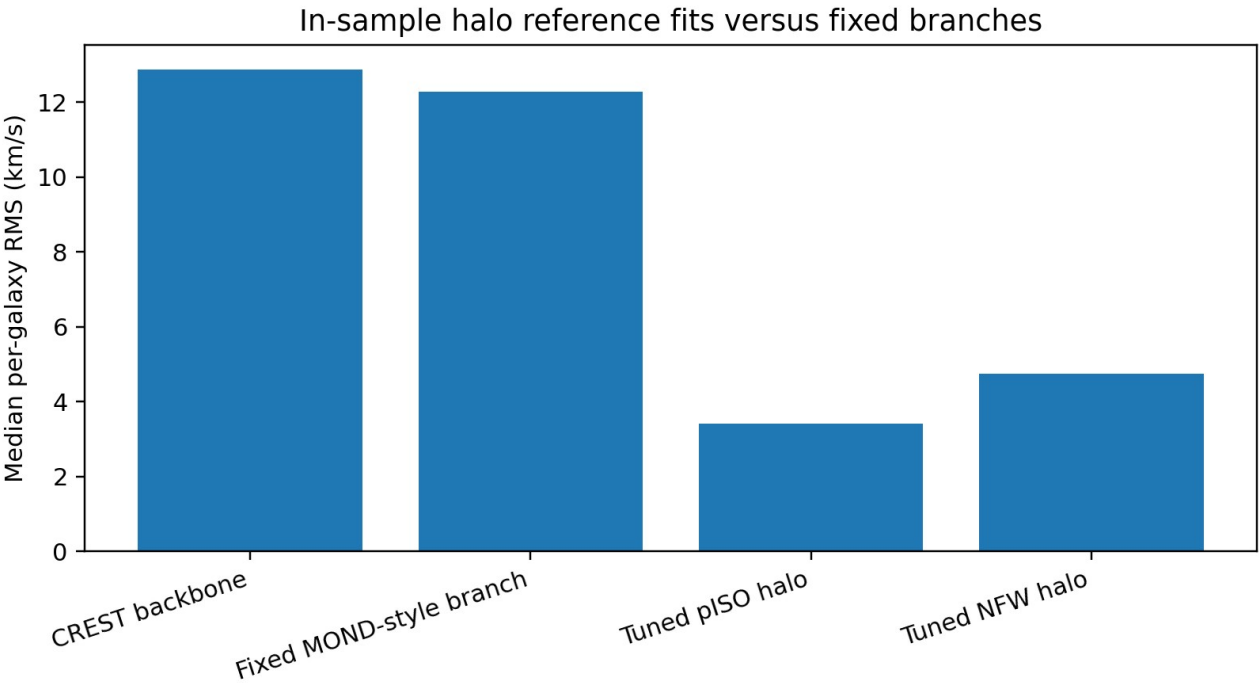


Figure 3. Tuned halo references remain much stronger as in-sample galaxy-by-galaxy fits.

11. Overlap cases and residual interpretation

A physically meaningful residual layer should eventually correlate with independent evidence for structural complexity, disequilibrium, non-circular flow, contamination, or branch thickness. The existing overlap cases are heterogeneous. DDO 168 and NGC 2403 show strong residual-layer gains. DDO 154 improves weakly. NGC 5055 worsens under the current shared residual term. This does not invalidate the whole branch, but it shows that a single blended residual regressor is too coarse.

Galaxy	Q	Backbone RMS	Huber RMS	Gain	External proxy summary
DDO 154	2	2.95	2.89	+2.0%	V/sigma=5.09; eps=0.197; Cglob=1.108
DDO 168	2	7.06	5.21	+26.2%	V/sigma=5.47; eps=0.183; Cglob=1.528
NGC 2403	1	8.57	6.22	+27.4%	Vrot=107.9; Cglob=0.208
NGC 5055	1	35.79	38.22	-6.8%	Vrot=169.9; Cglob=0.317

12. F-sector closure status

The F sector remains the most important unresolved empirical layer. Earlier cube/moment-map experiments found that raw map roughness and crude cube-derived moment structure did not close F. This is a useful negative result. It implies that F should not be defined as generic noise or roughness. It should be treated as irreversible or disequilibrium loading requiring true residual kinematic modeling.

- Bad F estimator: raw moment-map roughness alone.
- Better F estimator: residual velocity field after subtracting an axisymmetric or tilted-ring model.
- Stronger F estimator: 3DBarolo/BBarolo cube-level residual fields with beam, inclination, resolution, and non-circular-flow controls.
- Decisive Phase 1B target: demonstrate that an independently generated F estimator predicts where the residual layer helps or fails.

13. Required next closure: two-block residual law

The present residual layer is useful but blended. The next closure should split organized residual structure into two blocks: a structural-curvature block and a kinematic-disequilibrium block. The first corresponds mainly to the I sector; the second corresponds mainly to F. This split prevents the framework from treating every residual as the same kind of response.

Two-block residual decomposition:

$$\Delta_{\theta}(R) = \Delta_I(R) + \Delta_F(R)$$

Structural block:

$$\Delta_I(R) = f_I[\text{surface-density gradients, curvature, branch thickness, } Q_{\chi}]$$

Kinematic/disequilibrium block:

$$\Delta_F(R) = f_F[\text{residual velocity field, dispersion load, asymmetry, non-circular flow}]$$

Admissibility link:

$$\chi^2 + R_F = \eta I \exp(-\lambda_F F)$$

14. Regenerated reproducibility bundle

The missing Phase 1 assets are regenerated here as a reproducibility specification. The present compiled PDF includes the reconstructed tables and figures. The companion CSV assets created during compilation are:

phase1_rmse_table.csv, phase1_halo_reference_table.csv, and phase1_overlap_cases.csv. The executable notebook should be regenerated from the following skeleton.

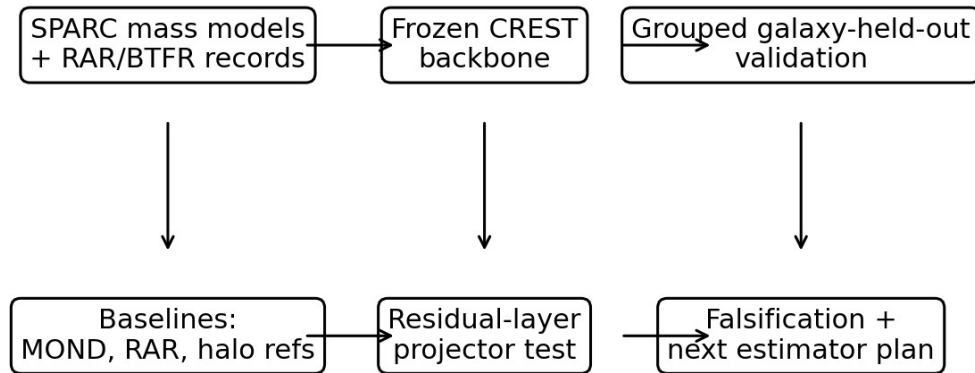


Figure 4. Phase 1 validation and regeneration workflow.

```

# CREST Phase 1 reproducibility skeleton
# Inputs: SPARC_Lelli2016c.mrt.txt, MassModels_Lelli2016c.mrt.txt, RAR.mrt.txt, BTFR_Lelli2019.mrt.txt

1. Load SPARC galaxy metadata and mass-model radial tables.
2. Construct  $g_{\text{obs}} = V_{\text{obs}}^2 / R$  after unit conversion.
3. Construct  $g_{\text{bar}}$  from gas, disk, and bulge contributions with fixed  $\text{Upsilon}_{\text{d}}$  and  $\text{Upsilon}_{\text{b}}$ .
4. Remove rows with invalid radii, missing velocities, missing uncertainties, or nonphysical accelerations.
5. Assign galaxy identity groups and Q quality flags.
6. Define frozen predictors:
    CREST0 =  $g_{\text{bar}} + \sqrt{a_0 * g_{\text{bar}}}$ 
    MOND_simple =  $0.5 * g_{\text{bar}} * (1 + \sqrt{1 + 4*a_0/g_{\text{bar}}})$ 
    RAR =  $g_{\text{bar}} / (1 - \exp(-\sqrt{g_{\text{bar}}/a_0}))$ 
7. Run grouped 5-fold cross-validation by galaxy identity.
8. Fit  $a_0$  on training galaxies only for MOND/RAR comparators when a fitted comparator is declared.
9. Train the shared CREST residual model using only allowed non-target-derived features.
10. Report held-out RMSE in  $\log_{10} g$  overall and by Q flag.
11. Run null radial shuffling to test whether improvement depends on organized radial structure.
12. Run ablations removing I features and F features separately.
13. Save result CSVs and figures with software versions and random seed.

```

15. Acceptance, partial-success, and failure criteria

The Phase 1 claim must be falsifiable. The following criteria should be published before additional re-fitting.

Outcome	Criterion	Interpretation
Strong success	Global no-retuning CREST residual projector beats fixed MOND/RAR under grouped galaxy-held-out validation and survives radial-shuffle/null tests.	Supports a genuine organized response layer.
Partial success	CREST improves only in specific Q flags, morphology regimes, or kinematic-complexity classes.	Supports a regime-dependent response layer; narrows the domain of validity.
Negative but useful	CREST does not improve held-out RMSE, but failures correlate with missing F/I	Estimator is inadequate; the framework requires a better projector.

	estimator limits.	
Failure of current branch	CREST improvement disappears under grouped splits, radial shuffling, non-target-derived features, or matched baselines.	Current galaxy branch should be rejected or rebuilt.
Failure against halo references	Matched effective-degree halo models beat CREST under the same held-out protocol.	CREST remains empirically subordinate unless it offers separate explanatory value.

16. Risk controls against overclaiming

The framework is unusually broad. Phase 1 must therefore be protected from speculative inflation. The following restrictions keep the public document credible.

- Do not claim that CREST has solved dark matter.
- Do not claim that CREST has defeated MOND in all senses; the fixed backbone does not beat the fixed MOND/RAR branches in the recorded benchmark.
- Do not compare tuned halo models against fixed global laws as if they were the same kind of test.
- Do not use target-derived $g_{\text{obs}} - g_{\text{bar}}$ features in the public predictive model without labeling them diagnostic.
- Do not extend the Phase 1 evidence to consciousness, free energy, time machines, or radical endpoints.
- Do not present raw map roughness as a closed F estimator.

17. Phase 1B execution checklist

The next operational step is to turn this technical document into a public reproducibility release. The release should contain one notebook, one data manifest, one result folder, one figures folder, and one PDF paper.

- Recover or download the four public SPARC/RAR/BTFR data files into a clean data/ folder.
- Create scripts/load_sparc.py with deterministic parsing and unit conversion.
- Create scripts/build_features.py for g_{bar} , g_{obs} , CREST0, MOND, RAR, I proxies, and allowed F proxies.
- Create scripts/run_grouped_cv.py with grouped 5-fold validation and fixed random seed.
- Create scripts/run_nulls.py for radial shuffle and label-shuffle null tests.
- Create results/phase1_main_results.csv and results/phase1_q_stratified.csv.
- Create figures/phase1_model_hierarchy.png and figures/phase1_quality_strata.png.
- Create README.md explaining how to reproduce the figures.
- Update this PDF from Phase 1A technical consolidation to Phase 1B executable validation.

18. External critique package

The correct outreach posture is modest and technical. The message should ask reviewers to attack the assumptions, not to endorse the worldview. A useful target reviewer is someone who works with rotation curves, radial acceleration relation tests, MOND, halo fitting, galaxy dynamics, or statistical model comparison.

Suggested critique request:

I am testing whether a response-field residual model captures organized structure in public SPARC rotation-curve data under grouped galaxy-held-out validation. I am not claiming to have solved dark matter. I would value criticism of the split protocol, baseline comparison, feature admissibility, and whether the residual layer is physically assigned rather than target-leaking.

19. Conclusion

Phase 1 should be framed as the first credibility engine of the larger framework. It is not the whole theory and not the radical endpoint program. The archived benchmark gives a disciplined mixed result: fixed CREST backbone alone is weaker than fixed MOND/RAR, the shared residual layer improves aggregate held-out performance, Q-stratified behavior suggests activation in messier systems, tuned halo references remain a major empirical gap, and F-sector closure requires true independent kinematic modeling. This is enough to justify a public technical paper and reproducibility release, but not enough to claim final astrophysical closure.

The next strongest move is to publish the Phase 1A document as a specification, regenerate the executable notebook and CSV outputs as Phase 1B, and then ask technically hostile reviewers to break it.

Appendix A. Regenerated asset manifest

File	Description
phase1_rmse_table.csv	Reconstructed main held-out RMSE table.
phase1_halo_reference_table.csv	Reconstructed median RMS halo-reference table.
phase1_overlap_cases.csv	Overlap cases linking residual-layer behavior to external kinematic diagnostics.
figure_1_full_rmse.png	Full RMSE model hierarchy plot.
figure_2_q_stratified_rmse.png	Q-stratified RMSE plot.
figure_3_halo_reference.png	Tuned halo reference comparison plot.
figure_4_phase1_workflow.png	Phase 1 reproducibility workflow diagram.

Appendix B. Minimal public repository structure

```
crest-phase1/  
  README.md  
  environment.yml  
  data/  
    SPARC_Lelli2016c.mrt.txt  
    MassModels_Lelli2016c.mrt.txt  
    RAR.mrt.txt  
    BTFR_Lelli2019.mrt.txt  
  scripts/  
    load_sparc.py  
    build_features.py  
    run_grouped_cv.py  
    run_nulls.py  
    make_figures.py  
  notebooks/  
    CREST_Phase1_Reproduction.ipynb  
  results/  
    phase1_main_results.csv  
    phase1_q_stratified.csv  
    phase1_halo_reference.csv  
    phase1_null_tests.csv  
  figures/  
    phase1_model_hierarchy.png  
    phase1_quality_strata.png  
    phase1_halo_reference.png  
  paper/  
    CREST_Phase1_Technical_Document.pdf
```

Appendix C. Source basis

Source	Use in this document
crest2_updated.pdf	Primary benchmark values, grouped held-out comparison, halo reference table, two-block residual motivation.
CREST_metric_response_SPARC_paper.pdf	SPARC-facing metric-response framing, notation summary, and bibliographic references.
CREST_IV_metric_response_projector.pdf	Projector discipline, degraded reserve law, and falsifiability language.
CREST_metric_F_whitepaper.pdf	F-sector framing and disequilibrium interpretation.
Aletheian_Invariant_Projection_Architecture.pdf	Invariant role grammar and cross-domain projector orientation.

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